Plan the electric car

antocciphysics | TEACHER KEY | CPA

Learning target: Use circuit and mechanical evidence to plan a controlled motor-car test.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 14; 3 course days later.*
Faster flux change or more turns increases emf magnitude. Stronger field/effective area changes are valid with suitable controls.

02. Explain a worked example (worked)
Circuit shows electrical connections; placement plan shows physical clearances. They answer different design questions.

03. Look and explain (model)
All four parts form one series loop; any break stops steady current.

04. Build the solution (practice)
2 m/s; actual trial entries remain blank until measured.

05. Use the idea (reason)
No; confounded changes. Keep battery and alignment matched while varying wheel choice, use repeated runs.

06. Choose a method (mixed) *Review: Unit 7, Day 5; 12 course days later.*
$Q=It=24$ C.

07. Exit check (exit)
Teacher circuit check, secured components, clear rotating parts; measure distance/time or other approved quantity.

08. STRETCH: Apply it another way (stretch)
Either side of motor can interrupt current; accessibility, insulation and wire routing guide placement.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 14; 3 course days later.*
Emf is 3 times as large. Same turn count and flux change per turn; compare average emf over each interval.

02. Explain a worked example (worked)
Circuit shows electrical connections; placement plan shows physical clearances. They answer different design questions.

03. Represent and interpret (model)
All four parts form one series loop; any break stops steady current.

04. Solve and justify (practice)
1.5 W; 9 J; .833 m/s.

05. Analyze the reasoning (reason)
No; confounded changes. Keep battery and alignment matched while varying wheel choice, use repeated runs.

06. Choose a method (mixed) *Review: Unit 7, Day 5; 12 course days later.*
Circuit response involves propagation of electromagnetic changes and local charge motion. Drift speed is not the propagation speed of the signal.

07. Exit check (exit)
Motor/wire heating, rolling friction, sound, wheel rotational energy and drag; accept relevant terms.

08. STRETCH: Apply it another way (stretch)
4.8 versus 14.4 W. Use brief teacher-approved observation, remove power promptly, avoid heat buildup and use specified current protection.